

EE4437 HW1

Infinite Quantum Well:

$$m_e = 0.067 m_0, \quad m_0 = 9.11 \times 10^{-31}$$

when $L_z = 5 \text{ nm}$,

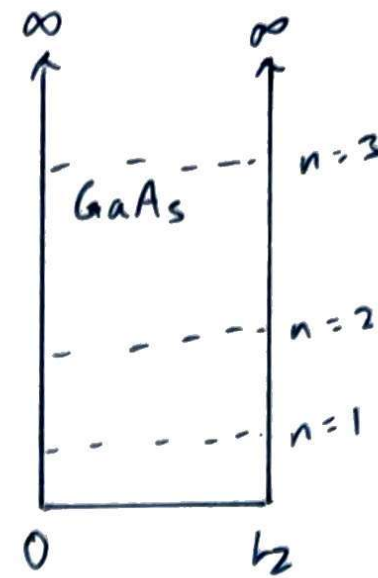
$$E_{n\kappa} = \frac{\hbar^2}{2m_e} \left(\frac{n\pi}{L_z} \right)^2 = n^2 \frac{\hbar^2}{2m_e} \left(\frac{\pi}{L_z} \right)^2$$

$$= n^2 \times 0.225 \text{ eV (3sf)}, \quad n = 1, 2, \dots$$

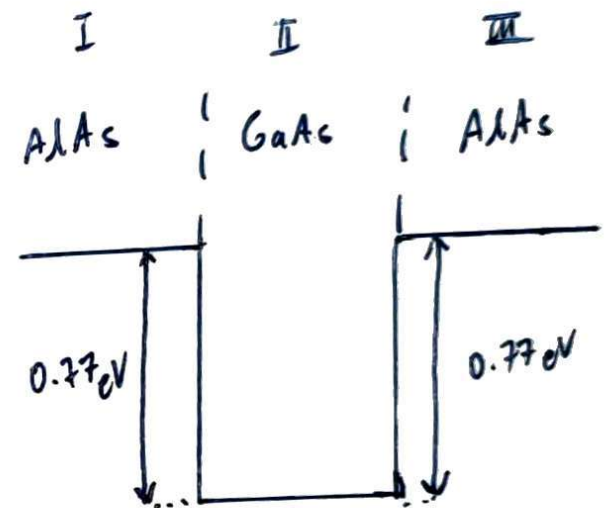
when $L_z = 10 \text{ nm}$,

$$E_{n\beta} = E_{n\alpha} \times \left(\frac{1}{2} \right)^2$$

$$= n^2 \times 0.0562 \text{ eV (3sf)}, \quad n = 1, 2, \dots$$



✓ this is indeed the same as if we calculated it from scratch



Finite Quantum Well:

m AlAs, $\kappa = 1$

GaAs, $\kappa = 0$

$$\Rightarrow \Delta E_c = 0.77 \text{ eV}$$

$$m_{e,Al} = (0.067 + 0.083)m_0 = 0.15m_0$$

$$m_{e,Ga} = 0.067m_0$$

let $E_{1\alpha}^{\infty} = 0.225 \text{ eV}$ when $L_2 = 5 \text{ nm}$,

$E_{1\beta}^{\infty} = 0.0562 \text{ eV}$ when $L_2 = 10 \text{ nm}$

let $\epsilon_{n\alpha} = \frac{E_{n\alpha}}{E_{1\alpha}^{\infty}}$, $\epsilon_{n\beta} = \frac{E_{n\beta}}{E_{1\beta}^{\infty}}$

let $V_{0\alpha} = \frac{V_0}{E_{1\alpha}^{\infty}}$, $V_{0\beta} = \frac{V_0}{E_{1\beta}^{\infty}}$

When $L_2 = 5 \text{ nm}$:

$V_{0\alpha} = \frac{\Delta E_0}{0.225} = 3.4222 \text{ (5sf)}$

with ϵ as x-axis, when plotting $y = \sqrt{V_0 - \epsilon}$, $y = \sqrt{\epsilon} \tan\left(\frac{\pi}{2}\sqrt{\epsilon}\right)$, $y = -\sqrt{\epsilon} \cot\left(\frac{\pi}{2}\sqrt{\epsilon}\right)$, we get intersections at $\epsilon = 0.54546$ and $\epsilon = 2.0588$

$\Rightarrow E_{1\alpha} = 0.54546 \times 0.225 \text{ eV} \quad E_{2\alpha} = 2.0588 \times 0.225 \text{ eV}$
 $= 0.123 \text{ eV (3sf)} \quad = 0.463 \text{ eV (3sf)}$

When $L_2 = 10 \text{ nm}$:

$V_{0\beta} = \frac{\Delta E_0}{0.0562} = 13.701$

as same procedure for previous case, we get intersections at

$\epsilon = 0.7261$, $\epsilon = 2.87924$, $\epsilon = 6.3637$, $\epsilon = 10.88646$

$\Rightarrow E_{1\beta} = 0.7261 \times 0.0562 \text{ eV}$
 $= 0.0408 \text{ eV (3sf)}$

$E_{2\beta} = 0.162 \text{ eV (3sf)}$

$E_{3\beta} = 0.357 \text{ eV (3sf)}$

$E_{4\beta} = 0.612 \text{ eV (3sf)}$

EE4437 HW1 Corrections

Infinite QW:

$$m_e = 0.067 m_0, \quad m_0 = 9.11 \times 10^{-31}$$

$$L_z = 5 \text{ nm}:$$

$$E_{n\alpha} = \frac{\hbar^2}{2m_e} \left(\frac{n\pi}{L_z} \right)^2$$
$$= n^2 \times 0.225 \text{ eV}, \quad n = 1, 2, 3, \dots$$

$$L_z = 10 \text{ nm}:$$

$$E_{n\beta} = E_{n\alpha} \times \left(\frac{L}{2} \right)^2 \quad \left. \begin{array}{l} \nearrow \\ \searrow \end{array} \right\} \text{3sf}$$
$$= n^2 \times 0.0562 \text{ eV}, \quad n = 1, 2, 3, \dots$$

Finite QW, same masses:

$$\text{m AlAs}, \quad x = 1$$

$$\text{GaAs}, \quad x = 0$$

$$\Rightarrow \Delta E_c = 0.77 \text{ eV}$$

$$m_b = 0.15 m_0$$

$$m_w = 0.067 m_0 \quad \leftarrow \text{not used for this case}$$

$$\text{let } E_{1\alpha}^\infty = 0.225 \text{ eV} \quad (L_z = 5 \text{ nm}), \quad \text{and} \quad E_{1\beta}^\infty = 0.0562 \text{ eV} \quad (L_z = 10 \text{ nm})$$

$$E_{n\alpha} = \frac{E}{E_{1\alpha}^\infty}, \quad E_{n\beta} = \frac{E}{E_{1\beta}^\infty}$$

$$V_{0\alpha} = \frac{V_0}{E_{1\alpha}^\infty}, \quad V_{0\beta} = \frac{V_0}{E_{1\beta}^\infty}$$

$$L_z = 5 \text{ nm}:$$

$$V_{0\alpha} = \frac{\Delta E_c}{0.225} = 3.4222 \quad (\text{5sf})$$

$$\text{w } E \text{ as } x\text{-axis, plot } y = V_0 - E, \quad y = \sqrt{E} \tan\left(\frac{\pi}{2} \sqrt{E}\right), \quad y = -\sqrt{E} \cot\left(\frac{\pi}{2} \sqrt{E}\right)$$

$$\Rightarrow \text{intersections @ } E_1 = 0.54546, \quad E_2 = 2.0581$$

$$\begin{aligned} \Rightarrow E_{1\alpha} &= E_1 \times E_{1\alpha}^\infty & E_{2\alpha} &= E_2 \times E_{1\alpha}^\infty \\ &= 0.123 \text{ eV} & &= 0.463 \text{ eV} \end{aligned}$$

$$L_z = 10 \text{ nm}:$$

$$V_{0\beta} = 13.701$$

:

$$E_1 = 0.7261, \quad E_2 = 2.87929, \quad E_3 = 6.3637, \quad E_4 = 10.88646$$

$$\Rightarrow E_{1\beta} = 0.0408 \text{ eV}, \quad E_{2\beta} = 0.162 \text{ eV}, \quad E_{3\beta} = 0.357 \text{ eV}, \quad E_{4\beta} = 0.612 \text{ eV}$$

Finite QW, different mults:

$$k = \sqrt{\frac{2m_w E}{\hbar^2}} = \left(\frac{\pi}{L_z}\right) \sqrt{\frac{E}{E_1^\infty}} = \left(\frac{\pi}{L_z}\right) \sqrt{E} \quad (\text{same as previous})$$

$$\alpha = \sqrt{\frac{2m_b (V_0 - E)}{\hbar^2}} = \sqrt{\frac{0.15}{0.067}} \left(\frac{\pi}{L_z}\right) \sqrt{V_0 - E}$$

added $\sqrt{\frac{0.15}{0.067}}$ term, to account for $m_b = \frac{0.15}{0.067} m_w$

$$\frac{\alpha}{k} = \sqrt{\frac{0.15}{0.067}} \sqrt{\frac{V_0 - E}{E}}$$

$$\frac{kL_z}{2} = \frac{\pi}{2} \sqrt{E}, \quad \frac{\alpha L_z}{2} = \sqrt{\frac{0.15}{0.067}} \left(\frac{\pi}{2}\right) \sqrt{V_0 - E}$$

$$\tan\left(\frac{\pi}{2} \sqrt{E}\right) = \sqrt{\frac{0.15}{0.067}} \sqrt{\frac{V_0 - E}{E}} \Rightarrow \left(\sqrt{\frac{0.15}{0.067}}\right)^{-1} \sqrt{E} \tan\left(\frac{\pi}{2} \sqrt{E}\right) = \sqrt{V_0 - E}$$

$$-\cot\left(\frac{\pi}{2} \sqrt{E}\right) = \sqrt{\frac{0.15}{0.067}} \sqrt{\frac{V_0 - E}{E}} \Rightarrow -\left(\sqrt{\frac{0.15}{0.067}}\right)^{-1} \sqrt{E} \cot\left(\frac{\pi}{2} \sqrt{E}\right) = \sqrt{V_0 - E}$$

$$\therefore \text{Find intersections of } y = \sqrt{V_0 - E} \text{ and } y = -\left(\sqrt{\frac{0.15}{0.067}}\right)^{-1} \sqrt{E} \cot\left(\frac{\pi}{2} \sqrt{E}\right)$$

$$y = \left(\sqrt{\frac{0.15}{0.067}}\right)^{-1} \sqrt{E} \tan\left(\frac{\pi}{2} \sqrt{E}\right)$$

w/ E as x -axis

$$L_z = 5 \text{ nm}$$

$$V_0 = 3.4222$$

$$\text{intersections @ } E_1 = 0.64315, \quad E_2 = 2.29533$$

$$\Rightarrow E_{1\alpha} = 0.14471 \text{ eV}, \quad E_{2\alpha} = 0.51645 \text{ eV}$$

$$L_z = 10 \text{ nm}$$

$$V_0 = 13.701$$

$$\text{intersections @ } E_1 = 0.80093, \quad E_2 = 3.15703, \quad E_3 = 6.88295, \quad E_4 = 11.40175$$

$$\Rightarrow E_{1\beta} = 0.045012 \text{ eV}, \quad E_{2\beta} = 0.17743 \text{ eV}, \quad E_{3\beta} = 0.38682 \text{ eV},$$

$$E_{4\beta} = 0.64078 \text{ eV}$$